



UNIVERSITY EXAMINATIONS: 2025/2026
EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE IN
INFORMATION SECURITY AND FORENSICS
BISF 3205/CSN 4204: VULNERABILITY AND PENETRATION TESTING
FULL TIME/PART TIME/DISTANCE LEARNING
ORDINARY EXAMINATION

DATE: DECEMBER, 2025

TIME: 2 HOURS

INSTRUCTIONS: Question One is Compulsory, Choose Two Other Questions

QUESTION ONE (20 MARKS) COMPULSORY

- a) Discuss the role of a sniffer on the testing machine that runs the scanning tool when conducting penetration testing. **(5 Marks)**
- b) Compare and contrast at least four penetration testing methodologies, highlighting their pros and cons and why you would prefer one methodology over another. **(10 Marks)**
- c) A cyber security incident has just happened in the organization you work for. You are the penetration tester in the Security Operations Centre (SOC) of your organization and you suspect that the security of your organization has been compromised. Discuss in detail how this incident will be handled. **(5 Marks)**

QUESTION TWO (15 MARKS)

- a) Using real world examples, discuss in detail the penetration testing deliverables. **(7 Marks)**
- b) Develop a detailed template for preparing the Rules of Engagement for your penetration testing team, highlighting the considerations you must make before this development. **(8 Marks)**

QUESTION THREE (15 MARKS)

You have been given a vulnerable machine but you cannot log into the machine. You are required to set up this machine in a virtual environment and exploit. Discuss in detail every step you follow in setting up, configuring and making ready for exploitation. Discuss how you would go about conducting your penetration testing engagement up to full exploitation of the vulnerable machine.

(15 Marks)

QUESTION FOUR (15 MARKS)

a) You run the nmap command to extract details about a target. What do the following within the nmap command mean?

(i) -n **(1 Mark)**

(ii) -sP **(1 Mark)**

(iii) -oA **(1 Mark)**

(iv) -sV **(1 Mark)**

(v) -A **(1 Mark)**

b) Develop a detailed Code of Ethics for a Penetration Tester **(10 Marks)**